

VANISH: Vanishing-Factor Ablation of Non-tangent Flows in Simplex and Hypercube

Learned flows leave bounded domains; a vanishing factor restores feasibility, not quality

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Framing the study

Problem. Continuous generative models on *bounded* domains. The training objective regresses v_θ onto an admissible target, but v_θ itself is an unconstrained map — nothing forces the *fitted* field to point back into the domain.

Theory assumes it does: the no-flux condition for confined flows requires boundary tangency [1] and calls it automatic for the *true* field. Practice silently repairs the *learned* one — simplex projection [2], clamping, dynamic thresholding [3].

Sample-level violations are known [4]; how often the per-step repair fires on real sequence data is not. We measure it, then ask whether fixing it helps.

Two modalities, both bounded, otherwise unrelated:

- ▶ **Modality 1** — biological sequences on the probability simplex Δ_K^N
- ▶ **Modality 2** — images on the unit box $[0, 1]^{784}$

The simplex is a $(K-1)$ -dimensional affine slice with Fisher–Rao structure; the box is a full-dimensional product of intervals. They share *only* boundedness — which is what makes the transfer a real test.

Modality 1 — three datasets. Length $N=128$; train on 12,000, score against 4,351 held-out sequences; unconditional generation:

- ▶ **DNA**, $K=4$ — 128-base windows of GRCh38 chr1:1–10 Mb [1]
- ▶ **Codon**, $K=64$ — the *same region*, 384-base windows read as trinucleotides (3-mer tokens, not reading-frame codons)
- ▶ **Protein**, $K=20$ — reviewed human SwissProt, tiled into 128-residue windows [2]

Held-out split **by protein** and by **100 kb genomic block**, deduplicated — no protein or region on both sides.

Representation. One-hot simplex *vertices*, decoded by $\arg \max$ (the denoiser baselines decode their final posterior, as published).

Modality 2. MNIST [3] scaled to $[0, 1]^{784}$, then mapped *strictly inside* the domain ($x \mapsto 0.98x + 0.01$) — otherwise the data sit on the boundary and the constrained parameterisation is degenerate rather than tested. Train on 20,000; hold out the next 5,000 as the Fréchet reference.

Conditioning. Unconditional; guidance (2b) acts at sampling time only.

Formulation. Linear conditional path between source and data,

$$x_t = (1 - t)x_0 + tx_1, \quad u_t = x_1 - x_0,$$

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1], x_0, x_1} \|v_\theta(x_t, t) - (x_1 - x_0)\|_2^2,$$

with x_0 uniform on the simplex per position [1].

Architecture and training. Dilated 1-D CNN, the standard backbone here: 8 residual blocks, kernel 5, dilation cycling 1, 2, 4, 8, 16, width 128, zero-initialised output, sinusoidal time features. Adam, lr 10^{-3} , batch 128, 8000 steps.

Model selection. Fixed step budget, *not* validation early stopping — identically for every arm, so no arm gets a selection advantage.

Sampling. Explicit Euler on a uniform grid at $\text{NFE} \in \{10, 50, 100\}$, $x \leftarrow x + h v_\theta$, then clamp and renormalise. **That clamp is the standard practice whose consequences this whole study measures.**

Formulation. Variance-preserving diffusion on the same simplex coordinates, cosine schedule [1]

$$\bar{\alpha}(s) = \cos^2\left(\frac{\pi}{2} \cdot \frac{s + 0.008}{1.008}\right), \quad x_s = \sqrt{\bar{\alpha}} x_1 + \sqrt{1 - \bar{\alpha}} \epsilon,$$

trained with $\mathcal{L} = \mathbb{E}\|\epsilon - \epsilon_\phi(x_s, s)\|^2$ and sampled by DDIM, $x_{s'} = \sqrt{\bar{\alpha}_{s'}} \hat{x}_1 + \sqrt{1 - \bar{\alpha}_{s'}} \epsilon_\phi$.

Architecture and training. *Identical* to the flow-matching baseline — architecture, parameter count, optimiser, learning rate, batch, steps, seeds, evaluation sample count and NFE grid. Only the generative process differs.

Model selection. Fixed step budget, as above.

Sampling. Deterministic DDIM [2] at $\text{NFE} \in \{10, 50, 100\}$, so NFE means the same thing in both families.

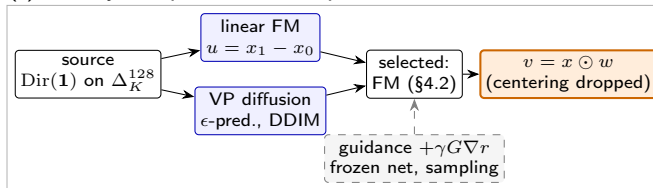
Metrics — deliberately oracle-free. We avoid learned evaluators on sequences, so no classifier's accuracy limits the metric:

- ▶ **step-violation rate** — fraction of raw Euler updates leaving the domain, computed *unclamped* in every arm, so the number means the same thing across parameterisations
- ▶ **x_0 -outside rate** — fraction of steps whose implied clean-sample estimate $x + (1 - t)v$ is out of domain
- ▶ **marginal KL** and **3-mer correlation** [1] — distributional quality, against a **no-model unigram sampler** as the floor
- ▶ **NFE** — sampling efficiency

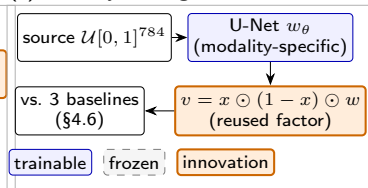
Fairness. One factor varies at a time; everything else is matched. **8 seeds** on Modality 1, 8 on Modality 2. Uncertainty is a paired bootstrap (20,000 resamples, paired on seed); an effect counts only if the 95% interval excludes zero; headline effects also get Student- t intervals and seed counts. 4,000 samples per cell (5,000 images).

Preregistered: the 4.2 run, held-out data and split, corrected baselines, Modality 2 rerun, real-data guidance. **Not:** the decomposition, the 3-mer endpoint.

(a) Modality 1: sequences on the simplex



(b) Modality 2: images in the box



The pipeline. Establish both baselines under one protocol $\rightarrow$ select a family (4.2) $\rightarrow$ show controllability (4.3) $\rightarrow$ measure the pathology $\rightarrow$ decompose the candidate fix and ablate it one piece at a time (4.4) $\rightarrow$ position against three recent external methods (4.5) $\rightarrow$ transfer the surviving mechanism to a second, unrelated bounded geometry (4.6).

Choosing the base model

Decision rule, fixed before running: lower marginal KL at NFE=10. **Confirmed on the real sequences** (dilated CNN, 8 seeds):

Dataset	FM, NFE 10	Diff., NFE 10	Diff. – FM, NFE 10	Diff. – FM, NFE 100	x_0 out (FM / Diff.)
DNA ($K=4$)	0.0061 $\pm$ 0.0043	0.7259 $\pm$ 0.4016	+0.720 [+0.447, +1.005]	+0.119 [+0.054, +0.194]	0.979 / 0.993
Protein ($K=20$)	0.3120 $\pm$ 0.0632	2.2958 $\pm$ 0.4275	+1.984 [+1.724, +2.265]	+0.301 [+0.220, +0.391]	1.000 / 1.000
Codon ($K=64$)	0.7483 $\pm$ 0.1314	1.7055 $\pm$ 0.6730	+0.957 [+0.570, +1.415]	+0.647 [+0.375, +0.991]	1.000 / 1.000

Selection. Flow matching, better on 7/8 seeds at every NFE on every dataset. The preregistered synthetic run agreed but used a small MLP and no real data — a disclosed deviation, and the reason for this confirmation.

The honest counterweight. Diffusion was untuned (harness sized for flow matching), so we claim the ordering, not the margin. On protein, 3-mer r at NFE=10 favours diffusion (+0.007 vs -0.167), and on DNA and codon a unigram sampler already reaches KL 0.002–0.019, so this metric mostly tests composition.

And the finding that set up the rest of the talk: the x_0 -outside rate is the same for both families — 0.979 vs 0.993 on DNA, 1.000 for both on codon. **The pathology is not a property of one objective.**

Guidance works — and what it costs

Setup. The frozen **innovation model** ($x \odot w_\theta$) on real sequences; drift $x \odot w_\theta + \gamma \nabla r$, then simplex projection (Euclidean rule), γ constant; reward $r = \sum_l \log \sum_{i \in S} x_{l,i}$ toward **GC content** (DNA) and **hydrophobic fraction** (protein). Nothing retrained; 8 seeds, 4,000 samples.

γ	DNA (GC)			Protein (hydrophobic)		
	Hit $\uparrow$	$r_{\text{top}} \uparrow$	Ham.	Hit $\uparrow$	$r_{\text{top}} \uparrow$	Ham.
0 (unguided)	0.09	0.59	0.75	0.03	0.60	0.94
0.03 (low)	0.30	0.92	0.74	0.55	0.74	0.94
0.3 (default)	1.00	0.65	0.53	1.00	0.53	0.84
3 (high)	1.00	0.55	0.44	1.00	0.50	0.72
<i>Held-out real</i>	0.09	—	0.75	0.10	—	0.94

Controllability is established: at $\gamma=0.3$ the hit rate (above the real 90th percentile) rises by $+0.908 [+0.870, +0.933]$ (DNA) and $+0.974 [+0.965, +0.981]$ (protein), with no violations and no extra runtime.

The cost is diversity and realism. Hamming falls by $-0.215 [-0.221, -0.211]$ / $-0.102 [-0.110, -0.096]$; strong guidance overshoots real high-property sequences. Post hoc, the sweet spot is small γ : similarity to the real top decile rises $0.59 \rightarrow 0.94$ on DNA at $\gamma=0.1$.

Table from results/guidance_real_*.jsonl; our own runs.

The innovation: which half of the fix works?

Motivation — the measured pathology. The learned field leaves the simplex on 0.71 of steps at $K=4$, 0.97 at $K=20$, 1.00 at $K=64$ (the rate counts any of $128K$ coordinates, so per coordinate it *falls*). On a synthetic $K=8$ model the rate is flat (0.21–0.22) over a $32\times$ step-size sweep — a property of the *fit*; on real DNA it still rises from 0.24 at NFE 10.

Specification. The replicator parameterisation [1] is $v = x \odot (w - \langle x, w \rangle)$ — and it bundles two distinct mechanisms:

- ▶ the factor x_i makes each component vanish on the face $x_i = 0$ (the flow cannot exit)
- ▶ the centering $\langle x, w \rangle$ gives $\sum_i v_i = 0$ (stay on the affine hull)

Not separated in flow matching; Wang & Kanamori [2] use the same face-vanishing factor for reflected score learning. We test the factor *alone*:

$$\text{mult-only } v = x \odot w, \quad \text{center-only } v = w - \bar{w},$$

against the base $v = w$ and the full form. Adds no parameters and no hyperparameters.

Testing the innovation: one piece at a time

Arm	3-mer r			step violation		
	$K=4$	$K=20$	$K=64$	$K=4$	$K=20$	$K=64$
base w	0.894	0.753	0.127	0.713	0.967	1.000
mult $x \odot w$	0.899	0.865	0.196	0.0000	0.0000	0.0000
center $w - \bar{w}$	0.910	0.777	0.120	0.721	0.967	1.000
full replicator	0.896	0.836	0.187	0.010	0.010	0.016

The multiplicative factor does the work. Mult-only improves over base by $+0.113$ [$+0.085, +0.139$] at $K=20$ and $+0.069$ [$+0.049, +0.089$] at $K=64$ on held-out data — 8/8 and 8/8 seeds; t -intervals agree. Violation is 0.0000 at $\text{NFE} \geq 50$; 8.7–10.0% at $\text{NFE} 10$, about one step in ten, consistent with the last.

Center-only is inert — non-significant at every K (the base target already sums to zero). The full replicator also helps, and **dropping its centering changes nothing** ($+0.009$ [$-0.010, +0.031$] at $K=64$).

The clamping confound is not ruled out. A gain from avoiding the clamp shrinks as NFE grows. At $K=20$ ours does ($+0.337 \rightarrow +0.113$); at $K=64$ it is not significant at $\text{NFE} 10$ and appears from $\text{NFE} 50$.

Method	3-mer $r \uparrow$ (NFE 100)			Step viol. $\downarrow$
	DNA ($K=4$)	Protein ($K=20$)	Codon ($K=64$)	$K=4/20/64$
Dirichlet FM	0.901 $\pm$ 0.062	0.906 $\pm$ 0.020	0.447 $\pm$ 0.020	0.00/0.00/0.00
Fisher-Flow	0.909 $\pm$ 0.049	0.886 $\pm$ 0.016	0.295 $\pm$ 0.013	0.00/0.00/0.00
Gumbel-Softmax FM	0.894 $\pm$ 0.052	0.705 $\pm$ 0.061	0.400 $\pm$ 0.070	0.00/0.00/0.00
Linear FM (internal base)	0.894 $\pm$ 0.088	0.753 $\pm$ 0.058	0.127 $\pm$ 0.023	0.71/0.97/1.00
Ours: $v = x \odot w$	0.899 $\pm$ 0.061	0.865 $\pm$ 0.027	0.196 $\pm$ 0.016	0.00/0.00/0.00
Reference: unigram sampler (no model)	-0.046 $\pm$ 0.025	0.900 $\pm$ 0.002	0.218 $\pm$ 0.002	—

Fair study. Each baseline keeps its own path, objective and decoding; backbone, budget and seeds matched; our re-implementations, nothing quoted.

We are not state of the art. At $K=4$ all are within noise. At $K=20, 64$ Dirichlet FM [1] leads us by +0.041 [+0.020, +0.071] and +0.250 [+0.230, +0.269], Fisher-Flow [2] by +0.020 [+0.001, +0.042] and +0.098 [+0.080, +0.114]; we beat Gumbel-Softmax FM [3] only at $K=20$. A **unigram sampler** scores 0.900 / 0.218: protein is near its composition floor; on codon ours is below it.

Feasibility does not separate them: all three are feasible in their own samplers.

The transferred method

What stays exactly the same. The mechanism is a parameterisation of the *velocity*, not an architecture — so it transfers verbatim in form. The invariant idea is *a factor that vanishes where the domain ends*:

$$\text{simplex: } v = x \odot w \quad \longrightarrow \quad \text{box: } v = x \odot (1 - x) \odot w,$$

vanishing at *both* faces of every interval. Still no added parameters, no added hyperparameters, same loss, same preregistered protocol.

What changes, and why it is not the mechanism. The box has no sum-to-one constraint, so there is no centering term to drop — **the box arm *is* the mult-only form**, which is precisely why Modality 2 tests the half of the innovation that survived Part 2.

Training. U-Net backbone (two down stages, mid block, symmetric up with skips, GroupNorm, SiLU, sinusoidal time embedding per block), Adam lr 2×10^{-4} , batch 128, 6000 steps, 8 seeds, 5,000 evaluation samples (0.60M parameters).

Sampling. Euler at $\text{NFE} \in \{10, 50, 100\}$; violations counted on the *unclamped* update, exactly as in Modality 1.

Method	FD ↓	Coverage ↑	Density ↑	FD@10 ↓	Step viol. ↓
Reflection sampler	192.0 ± 143.5	0.624	0.478	540.3	0.1401
Mirror-map FM	52.5 ± 23.9	0.685	0.480	98.9	0.0000
Dyn. thresholding on x_t	195.5 ± 150.3	0.623	0.480	540.8	0.1604
Flow matching (internal base)	193.9 ± 145.6	0.623	0.478	545.3	0.1602
Ours: $v = x(1 - x) \odot w$	151.0 ± 47.4	0.410	0.302	269.6	0.0000

Fair study. Matched backbone, optimiser, schedule, batch, steps, seeds and evaluation count across all five arms. The three external arms are **our flow-matching re-implementations** [1,2,3]: reflection and thresholding are sampler rules on the base network, mirror trains in logit space — none is the published diffusion model. Quality is Fréchet distance in the feature space of a classifier we train ourselves, and **we report its test accuracy** (96.2%; one classifier, shared by every arm) because our critique of this field’s metrics applies to ours too. Diversity is coverage [4].

Fair discussion. See the following slide — the result is mixed, and the parts that go against us are on it.

Does the innovation transfer?

Direction and magnitude, compared across modalities.

- ▶ **Feasibility transfers.** Step violation is 0.0000 on the box at NFE 100, as on the simplex. Same mechanism, unrelated geometry — this part is unambiguous.
- ▶ **Quality does *not* transfer.** Against the internal base, $+42.9 [-41.8, +144.2]$ over 8 seeds — **contains zero**, so on our preregistered rule the transfer claim is **not supported**. Mirror-map FM beats us significantly ($-98.5 [-136.7, -65.6]$); we lose coverage and density on 8 and 8 of 8 seeds; reflection and thresholding act as plain clamping.
- ▶ **The mean rests on one seed.** Ours wins FD on 4 of 8 seeds; drop seed 0 and $+42.9$ becomes $+2.8$. Its spread is lower (s.d. 47.4 vs 145.6; exploratory, untested), and FD at NFE 10 is lower on 8 seeds — a gain that vanishes as steps shrink.
- ▶ **We refuted our own explanation of the cost.** We had said image data are *interior*, so the factor is free there. Measuring it: only 10.6% of MNIST pixels are interior, and $x(1-x)$ is throttled below a tenth of its maximum on 88.4% of them. **Both** modalities are boundary-concentrated; the cost is paid wherever data concentrate at the boundary.

Verdict. *Feasibility* transfers, for the same reason, in both geometries. *Quality* does not — the transfer claim fails its own preregistered test.

Limitations and failure modes

1. **The factor still overshoots at few steps.** At NFE 10, Euler exits on 8.7–10.0% of steps, and at $K=64$ it gives no gain there.
2. **Constraint satisfaction does not buy quality.** Every strong method is feasible — Dirichlet and Gumbel by vanishing mixtures, Fisher-Flow and mirror by coordinates — and they still differ widely; on protein no arm beats composition by much, and on codon ours is below it.
3. **On images we lose both fidelity and diversity:** density and coverage are lower on 8 and 8 of 8 seeds, and samples are visibly blotchier.
4. **High seed variance on Modality 2.** Three of five arms swing by almost an order of magnitude between seeds, which is why the seed count was raised $3 \rightarrow 8$ by dated amendment before reading the remaining seeds.
5. **Six interventions were falsified**, and we report them: the Modality 2 quality transfer, a soft tangency penalty, a convex-combination integrator (a no-op), a $(1 - t)^2$ loss reweighting that was *harmful* (-0.273 [$-0.528, -0.049$] vs. base; the t -interval includes zero), the endpoint parameterisation and the centering term.
6. **Scale.** All models are small (≈ 0.6 M parameters); a larger backbone was tried only on superseded data.

Back to the motivation. The tangency premise confined-flow theory assumes [1] does not carry over to *learned* fields: violation is the norm (0.71–1.00 of steps) in both generative families — but **every strong method is feasible**, by mixtures or by coordinates.

Mechanism. Of the two pieces bundled in the replicator form, only the multiplicative factor does anything; the centering term is inert, so **it can be dropped**. The likeliest reading is an inductive bias, not the constraint; part of the $K=20$ gain may be avoided clamping.

The narrowest claim we can defend. A vanishing factor removes violation (at $\text{NFE} \geq 50$) at no parameter cost and improves a linear flow at $K \geq 20$; its centering is unnecessary; it does not close the gap to recent methods, and on images it transfers feasibility but not quality.

Next experiment. The factor at a larger backbone on the held-out split — does its gain survive scale?

- ▶ Albergo, M.S., Boffi, N.M., Vanden-Eijnden, E. Stochastic interpolants: a unifying framework for flows and diffusions. *JMLR* 26(209), 2025.
- ▶ Boll, B., Gonzalez-Alvarado, D., Petra, S., Schnörr, C. Generative assignment flows for representing and learning joint distributions of discrete data. *arXiv:2406.04527*, 2024.
- ▶ Cont, R. Reflected diffusions, no-flux continuity equations and confined Lagrangian flows in bounded domains. *arXiv:2607.28344*, 2026.
- ▶ Davis, O., Kessler, S., Petrache, M., Ceylan, İ.İ., Bronstein, M., Bose, A.J. Fisher flow matching for generative modeling over discrete data. *NeurIPS* 37, 2024.
- ▶ Eijkelboom, F., Bartosh, G., Naesseth, C.A., Welling, M., van de Meent, J.-W. Variational flow matching for graph generation. *NeurIPS*, 2024.
- ▶ Ensembl. Ensembl 2023. *Nucleic Acids Research* 51(D1), 2023.
- ▶ Fishman, N., Klarner, L., De Bortoli, V., Mathieu, E., Hutchinson, M. Diffusion models for constrained domains. *TMLR*, 2023.
- ▶ Ho, J., Salimans, T. Classifier-free diffusion guidance. *arXiv:2207.12598*, 2022.
- ▶ Karras, T., Aittala, M., Aila, T., Laine, S. Elucidating the design space of diffusion-based generative models. *NeurIPS* 35, 2022.

- ▶ LeCun, Y., Bottou, L., Bengio, Y., Haffner, P. Gradient-based learning applied to document recognition. *Proc. IEEE* 86(11), 1998.
- ▶ Lipman, Y., Chen, R.T.Q., Ben-Hamu, H., Nickel, M., Le, M. Flow matching for generative modeling. *arXiv:2210.02747*, 2022.
- ▶ Liu, G.-H., Chen, T., Theodorou, E.A., Tao, M. Mirror diffusion models for constrained and watermarked generation. *NeurIPS*, 2023.
- ▶ Lou, A., Ermon, S. Reflected diffusion models. *ICML*, 2023.
- ▶ Naeem, M.F., Oh, S.J., Uh, Y., Choi, Y., Yoo, J. Reliable fidelity and diversity metrics for generative models. *ICML*, 2020.
- ▶ Nichol, A.Q., Dhariwal, P. Improved denoising diffusion probabilistic models. *ICML*, 2021.
- ▶ Saharia, C., et al. Photorealistic text-to-image diffusion models with deep language understanding (Imagen). *NeurIPS* 35, 2022.
- ▶ Song, J., Meng, C., Ermon, S. Denoising diffusion implicit models. *ICLR*, 2021.
- ▶ Stärk, H., Jing, B., Wang, C., Corso, G., Berger, B., Barzilay, R., Jaakkola, T. Dirichlet flow matching with applications to DNA sequence design. *ICML*, 2024.

- ▶ Tang, S., Zhang, Y., Tong, A., Chatterjee, P. Gumbel-softmax flow matching with straight-through guidance. *arXiv:2503.17361*, 2025.
- ▶ UniProt Consortium. UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Research* 51(D1), 2023.
- ▶ Wang, Z., Kanamori, T. Should the boundary term be learned in reflected diffusion? Conormal trace and reflection masking. *arXiv:2608.03469*, 2026.
- ▶ Wang, C., Uehara, M., He, Y., et al. Fine-tuning discrete diffusion models via reward optimization (DRAKES). *ICLR*, 2025.
- ▶ Wang, W., Carreira-Perpiñán, M.Á. Projection onto the probability simplex. *arXiv:1309.1541*, 2013.
- ▶ Xie, T., Zhu, Y., Yu, L., Yang, T., Cheng, Z., Zhang, S., Zhang, X., Zhang, C. Reflected flow matching. *ICML*, 2024.
- ▶ Zern, A., Schnörr, C. On the correspondence between replicator dynamics and assignment flows. *SSVM*, 2021.